

Small-Molecule Disruption of the IL-23 / IL-23 Receptor Interaction for Psoriasis: An Evidence-Based Target Assessment

PlayMolecule AI

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Abstract

Blocking the IL-23 / IL-23 receptor (IL23R) interaction is the best-validated deep-clearance axis in moderate-to-severe plaque psoriasis: anti-p19 antibodies deliver week-16 PASI90 rates of 73–75%, and the IL-23R R381Q loss-of-function variant in human genetics confirms that attenuated IL-23 signaling protects against psoriasis. The historical objection to this target — that a cytokine–receptor interface cannot be drugged by anything other than an injectable antibody — has now fallen: icotrokinra, an oral macrocyclic peptide that binds IL23R, beat deucravacitinib in Phase 3 (week-16 PASI90 approximately 55–57%) and its first approval has been reported in 2026. The open frontier is a true small molecule. In 2025, an induced orthosteric pocket on the IL-23 p19 subunit was solved in complex with a cyclic peptide (8UUI), together with a fluorescence-polarization probe and a pilot fragment screen — the first credible small-molecule entry point on the cytokine side of the interface. We review the biology and clinical validation of IL-23 blockade, the structural landscape of the p19:IL23R interface and the epitopes of competing antibodies, and we lay out a strategy taxonomy, a target product profile, the competitive landscape, and a staged preclinical plan built around the induced pocket, DNA-encoded and display libraries, and mixed-solvent MD cryptic-pocket mapping, with a peptide/macrocycle fallback validated by the icotrokinra precedent.

Executive summary

- **The target is clinically and genetically validated.** IL-23 p19 antibodies achieve week-16 PASI90 in approximately 62–75% of patients, and the human *IL23R* R381Q loss-of-function variant protects carriers from psoriasis — human-genetics de-risking few targets enjoy [2, 8]. IL-23-selective (p19) blockade is also IBD-safe, unlike IL-17A or dual p40 blockade.
- **The interaction is orally druggable — the modality excuse is gone.** Icotrokinra (JNJ-2113), an oral macrocyclic peptide antagonist of IL23R, achieved week-16 PASI90 of approximately 55%/57% in the two pivotal Phase 3 trials and was superior to the oral TYK2 inhibitor deucravacitinib [6]; its first approval (as Icotyde) was reported in 2026 at approval-press level (Section 3). Any new program must now benchmark against that number, not against TYK2 orals.

- **A small-molecule entry point now exists on the cytokine side.** A 2025 structural study solved [8UUU](#): an induced, cryptic orthosteric pocket on p19, wedged between the CD loop and helix D, occupied by a cyclic peptide with low-nanomolar SAR that sterically blocks the IL23R face; a fragment screen against the pocket has already been run [\[10\]](#) — the concrete starting point for fragment-, DEL- and FEP-driven discovery.
- **The program risk is potency and ligandability, not biology.** The p19:IL23R interface is small ($\sim 900 \text{ \AA}^2$) and flat, with the binding energy concentrated in a hotspot cluster (Trp156 on p19 is the standout); classical cytokine interfaces have defeated small-molecule efforts when pockets were absent. The plan below therefore kills fast on ligandability and keeps an icotrokinra-like peptide/macrocyclic as the validated fallback.

1 Introduction and target rationale

Psoriasis is a chronic immune-mediated skin disease driven by a self-amplifying loop between myeloid antigen-presenting cells, type-17 T cells and keratinocytes [\[2\]](#). The cytokine IL-23 sits at the top of this loop: it maintains and expands the pathogenic IL-17-producing compartment, and blocking it produces the deepest and most durable skin clearance available in dermatology. IL-23 is a heterodimer of the IL-23-specific p19 subunit ([IL23A](#), [UniProt Q9NPF7](#)) and the shared p40 subunit (IL12B); it signals through a receptor pair composed of the specific IL-23R ([UniProt Q5VWK5](#)) and the shared IL-12R β 1, activating JAK2/TYK2 and predominantly STAT3 [\[8, 12\]](#).

Three facts make the p19:IL23R protein–protein interaction (PPI) an attractive small-molecule target rather than merely an antibody target. First, human genetics: the R381Q loss-of-function variant of IL23R protects heterozygous carriers from psoriasis, IBD and ankylosing spondylitis, with a safety signal that is quantifiable rather than theoretical ([Section 8](#)). Second, psoriasis is a large, adherence-sensitive market in which an oral systemic is strongly favored at entry. Third, the interaction has now been drugged from both sides: from the receptor side by icotrokinra, an oral macrocyclic peptide whose first approval was reported in 2026 ([Section 3](#)), and from the cytokine side by the discovery of an induced orthosteric pocket on p19 bound by an inhibitory cyclic peptide [\[10\]](#) — the first proves the market, the second defines a structural foothold for medicinal chemistry.

This report assesses small-molecule disruption of the p19:IL23R interaction as a psoriasis therapeutic strategy. [Section 2](#) summarizes IL-23 axis biology and clinical validation; [Section 3](#) reviews the therapeutic landscape and the icotrokinra precedent; [Section 4](#) analyzes the structural landscape of the interface, antibody epitopes, and druggability; [Section 5](#) develops a strategy taxonomy with precedents; [Section 6](#) and [Section 7](#) present a target product profile and the competitive landscape; [Section 8](#) discusses safety and translational risks; and [Section 9](#) proposes a staged preclinical plan.

2 Biology of the IL-23 axis and validation in psoriasis

2.1 Pathway architecture

IL-23 is a ~ 55 kDa heterodimeric class-II cytokine (p19 + p40) that engages a heteromeric receptor: IL-23R, whose extracellular region comprises fibronectin-III-like domains, and IL-12R β 1, shared with IL-12. JAK2 associates with IL-23R and TYK2 with IL-12R β 1; ligand engagement activates the JAK2/TYK2 pair and predominantly STAT3, driving transcription of IL-17A, IL-17F, IL-22 and GM-CSF [\[8\]](#). The full quaternary signaling complex is now solved at the atomic level [\[7\]](#): IL-23R D1 binds p19 at the IL-23-specific orthosteric site, while IL-12R β 1 directly engages the shared p40 — a “gateway” architecture that underlies the divergent roles of IL-12 and IL-23 on T versus NK cells. In living cells, IL-23 binds IL-12R β 1 with ~ 7 -fold higher affinity than IL-23R alone, but heteromeric

receptor assembly raises affinity more than 1,000-fold to $K_d \approx 27$ pM, with a conformational change in the N-terminal receptor domains on binding — a binding-induced assembly mechanism that is attractive for PPI disruption [9].

In psoriasis, IL-23 (from myeloid dendritic cells and macrophages) maintains pathogenic IL-17-producing Th17, Tc17, $\gamma\delta$ T cells and ILC3s infiltrating plaques; keratinocytes respond with antimicrobial peptides, chemokines and hyperproliferation, feeding the loop forward [2]. Acting upstream of the terminal IL-17/IL-22 effectors, IL-23 blockade extinguishes the supply line of multiple type-17 subsets at once — and, unlike IL-17A blockade, does not blunt protective mucosal IL-17 outputs: the mechanistic reading of the very deep clearance and clean IBD behavior of anti-p19 antibodies.

2.2 Human genetics as target validation

The strongest validation is human: the IL23R R381Q variant (rs11209026), a loss-of-function allele that reduces STAT3 signaling and receptor stability, protects heterozygous carriers from psoriasis, IBD and ankylosing spondylitis, and additional rare *IL23R* mutations associate with decreased psoriasis risk (*Exp Dermatol* 2024, PMID 39306854). Notably, R381Q lies in the cytoplasmic tail of the receptor, outside every extracellular structure in the PDB — a caution that the protective biology cannot be addressed structurally, only functionally. Homozygosity for R381Q associates with increased susceptibility to chronic mucocutaneous candidiasis (*Eur J Immunol* 2025, PMID 40685773), quantifying the antifungal-immunity cost of complete IL-23 loss (Section 8).

2.3 Clinical validation: the antibody benchmarks

Week-16 PASI90 response rates, the de-facto benchmark for deep clearance, are summarized in Fig. 1. Risankizumab achieved PASI90 75.3% in UltIMMa-1 versus 42.0% for ustekinumab (PMID 30097359); guselkumab achieved 73.3% in VOYAGE-1, superior to adalimumab through week 48 (PMID 28057360); tildrakizumab reached approximately 62% in reSURFACE-1 (PMID 28596043); ustekinumab, the dual IL-12/IL-23 p40 antibody, reached PASI75 67.1% in PHOENIX-1 (PMID 18486739). At the top of the class, the IL-17A/F antibody bimekizumab reaches approximately 91% PASI90 (approximate; ranked first in recent network meta-analyses of PASI90; *Cureus* 2026, PMID 42181314). All cross-trial comparisons in Fig. 1 are informal — trials differ in population, background therapy and scoring — and the values are approximate. They nevertheless establish the efficacy bar: *week-16 PASI90 in the 55–75% band is what an IL-23-axis systemic must approach.*

3 Therapeutic landscape: antibodies, orals, and the icotrokinra precedent

The psoriasis armamentarium is tiered (Section 7 gives the full landscape). Systemic orals comprise classical immunosuppressants, the PDE4 inhibitor apremilast (PASI90 approximately 7%), the allosteric TYK2 inhibitor deucravacitinib (POETYK PSO-1: PASI75 58.4%, PASI90 approximately 27–29%, approximate; PMID 35820547), and the JAK inhibitors, which carry MACE/VTE boxed-warning class liabilities. Biologics span TNF- α (adalimumab, PASI90 approximately 29%), IL-17A/F (secukinumab $\sim$ 59%; bimekizumab $\sim$ 91%) and the IL-23 p19 antibodies (guselkumab 73%, risankizumab 75%, tildrakizumab $\sim$ 62%; approximate, cross-trial informal). Ustekinumab, the only approved p40 antibody, blocks both IL-12 and IL-23 and now sits below the p19 class.

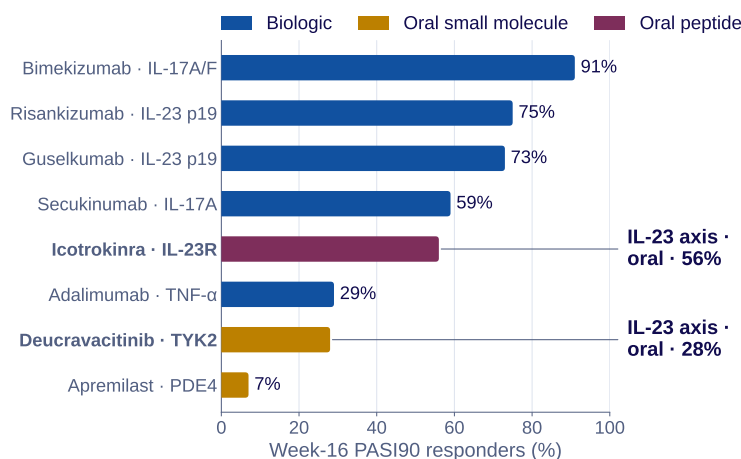


Figure 1. Week-16 PASI90 response rates across key benchmarks in moderate-to-severe plaque psoriasis (bars coloured by modality; each bar labelled with drug and mechanism). All values are approximate and the cross-trial comparison is informal: trials differ in population, endpoints and background therapy. Ictrokinra (56%) is the mean of the two pivotal Phase 3 ICONIC-ADVANCE trials; risankizumab (75%) is the lower end of the 75–84% range reported across trials; deucravacitinib is an allosteric TYK2 inhibitor, grouped here at program level rather than by IL-23R binding; bimekizumab (IL-17A/F) is included as the current class ceiling. The two IL-23-axis orals (ictrakinra, IL-23R; deucravacitinib, TYK2) are bolded. Compiled from the pivotal trials and network meta-analyses cited in the text.

The oral IL-23R precedent. Ictrokinra (JNJ-2113, ex-PN-235; Protagonist/Janssen) is a first-in-class *orally delivered macrocyclic peptide* that selectively binds IL23R and blocks IL-23 signaling without touching IL-12 [6]. In the pivotal Phase 3 ICONIC-ADVANCE 1 and 2 trials, week-16 PASI90 was 55%/57% versus 4%/1% for placebo, with superiority over deucravacitinib, and adverse-event rates similar to placebo (48% vs 57%) ([Lancet 2025, PMID 40976249](#)); long-term durability through 52 weeks was shown in the extension program. Its first approval as Ictotide for plaque psoriasis was reported in 2026 in first-approval sources [1, 5]; we treat that approval as established at the level of those reports — the precise approval date and label wording were not independently verified here, and Phase 2b topline figures quoted in the industry press (PASI75 approximately 80%, PASI90 approximately 50% at 240 mg once daily) are press-release-level and should be re-anchored to peer-reviewed sources before external use.

Two consequences follow. First, *the efficacy bar for an oral IL-23-axis agent is now an IL-23R antagonist, not a TYK2 inhibitor*: a new compound must clear approximately 55% week-16 PASI90 decisively ([Section 6](#)). Second, ictrakinra validates the pharmacology — and the safety — of blocking this interaction systemically, so a small molecule needs only to match, not discover, the clinical concept. What remains open is modality: a classical small molecule would offer cost-of-goods, stability and formulation advantages — if the interface can be liganded.

4 Structural landscape of the p19:IL23R interaction

4.1 Complex architecture

[Figure 2](#) shows the quaternary signaling complex: p19 (dark blue) and the shared p40 (gold) engage IL-23R (light blue) through its D1 domain and IL-12Rβ1 (plum) through p40 [7]. The IL-23-specific drug target is the p19:IL23R(D1) contact; the p40:IL-12Rβ1 leg is shared with IL-12 and must be avoided by a selective antagonist. This architecture has a direct selectivity consequence: *site II is absent from p19* — IL-12Rβ1 binds p40, not p19 — so any inhibitor aimed at the p19 surface is

inherently IL-23-selective and cannot perturb IL-12 signaling through the shared receptor subunit.

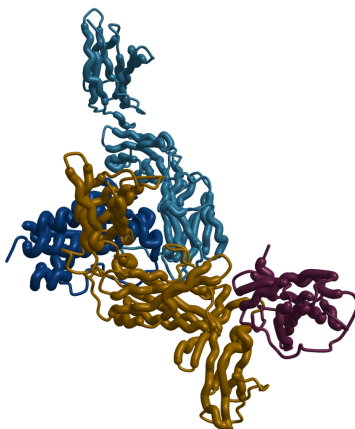


Figure 2. Quaternary IL-23 signaling complex ([6WDQ](#); 3.4 Å crystal structure of IL-23 with the extracellular domains of IL-23R and IL-12R β 1). IL-23 p19 in dark blue, the shared p40 subunit in gold, IL-23R in light blue, IL-12R β 1 in plum. IL-23R D1 binds the p19 orthosteric face; IL-12R β 1 binds p40 (site II is absent from p19), the gateway architecture that makes p19-targeted inhibitors inherently IL-23 selective. Rendered from [6WDQ](#); see Glassman et al. [7].

4.2 The p19:IL23R interface

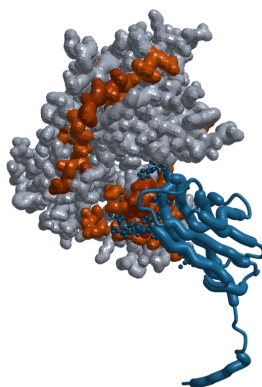


Figure 3. The p19:IL23R interface. p19 is shown as a white/gray surface with its IL23R-contact residues in orange (UniProt [Q9NPF7](#) numbering); IL-23R domain D1 is drawn as a blue cartoon. The contact patch spans the AB loop (54–60), V68, L103 and the N-terminal segment of helix D (154–168), including the Trp156 hotspot; the receptor contacts p19 only through D1. Rendered from the IL-23:IL23R complex ([5MZV](#); 2.8 Å).

The p19:IL23R interface is unusually small and flat for a cytokine–receptor pair: approximately 900 Å² involving 51 residues, with the receptor contacting the cytokine only via D1 ([Table 1](#) and [Fig. 3](#)). On p19, the contacts cluster in the AB loop (54–60), V68, L103, and the N-terminal segment of helix D (Q154/P155/W156/Q157, L159–L161, K164, I165, S168); on IL-23R they cluster in three D1 segments (G24–S31, I56, S98–T102, H108–D118). Trp156 (p19) stacks on IL-23R Gly116 and is the dominant hotspot; the helix-D N-terminus undergoes an α -helix to 3_{10} switch on binding, and the AB loop orders on binding — both induced-fit features that concentrate binding energy. A further structural constraint: IL23R Asn29 carries a glycan that is incompatible with the interface

Table 1. Residues defining the p19:IL23R interface. p19 residues are given in UniProt [Q9NPF7](#) numbering (mature p19 = UniProt – 19); IL-23R residues in UniProt [Q5VWK5](#) numbering. The interface spans approximately 900 Å² and 51 residues; the receptor contacts the cytokine only through domain D1 (residues 24–121). Compiled from the crystal structures [5MZV](#) and [6WDQ](#).

p19 structural element	p19 residues	IL-23R D1 partners
AB loop (54–60)	D55, L56, R57, E58, E59, G60	G24–S31 loop
Central bundle	V68, L103	I56, S98–T102
Helix-D N-terminus	Q154, P155, W156 (hotspot; stacks on IL-23R G116), Q157	H108–D118
Helix-D lower edge	L159–L161, K164, I165, S168	H108–D118 (lower rim)
Binding-induced change	Helix-D N-term $\alpha \rightarrow 3_{10}$ switch; AB-loop ordering	N29 glycan incompatible with interface

geometry, and p19 itself carries *no* N-glycosylation sequons (p40 carries them) — convenient for bacterial expression of a screening target.

Because IL-12R β 1 engages p40 rather than p19, p19-side binders face only the p40-mediated IL-12 cross-talk risk if they spill onto p40; the p19 contact residues listed in [Table 1](#) are IL-23-specific surface.

4.3 PDB structure inventory

[Table 2](#) inventories the verified IL-23 / IL-23R structure set. Two entries matter most for small-molecule work: [8UUI](#), the cyclic peptide 23-446 bound in an induced orthosteric pocket on p19 at 2.43 Å [10], and [6WDQ/8OE4](#), the full signaling complexes. It is equally important what is absent: no co-structures exist for any marketed p19 antibody (risankizumab, guselkumab, tildrakizumab, mirikizumab); their epitopes are known only from HDX-MS mapping ([Section 4.4](#)). Only unliganded Fab fragments (e.g. guselkumab Fab [4M6M/4M6N](#), CNTO4088 Fab [3I2C](#)) are deposited. Several early database entries sometimes attributed to IL-23 ([3DUW](#), [4GDK](#), [5MZW](#), [7PX5](#)) were verified *not* to be IL-23 structures and are excluded.

4.4 Antibody and scaffold epitopes, and overlap with the receptor

Structural co-structures exist for experimental binders, not for the marketed antibodies. Nanobody [37D5](#) ([4GRW](#)) contacts p19 residues 43–49, 112–123 and 156–163 and overlaps the IL-23R epitope including Trp156; Fab [7G10](#) ([3D85](#)) covers p19 101–158 including the Trp156 edge; briakinumab ([5NJD](#)) binds the p40 D1–D2 junction, tilted toward p19 (105–143, 32–40) [7]. The marketed p19 antibodies (risankizumab, guselkumab, tildrakizumab, mirikizumab) have *no* co-structures with IL-23; their epitopes are known only from HDX-MS mapping in a 2024 study (Daniele et al., *JID Innovations*) — a method-derived mapping that should be read with that caveat. On that mapping, epitope solvent-accessible surface area ranks risankizumab ($\sim 2,400$ Å²) > guselkumab ($\sim 2,240$ Å²) > ustekinumab ($\sim 1,390$ Å², binds p40) > tildrakizumab ($\sim 1,290$ Å²), and *larger epitope size associates with higher week-16 PASI90* ($R^2 \approx 0.997$ across $n=4$ antibodies — an association in four data points, not a dose-response relationship, and the HDX-derived SASA values are approximate). Guselkumab overlaps most of a 24-residue IL-23R-epitope definition (14 of 24 residues), risankizumab 2 of 24, tildrakizumab 0; all p19 antibodies share a 10-residue overlap outside the receptor epitope (S114–Q123, the CD loop). [Figure 5](#) shows the association; ustekinumab is included as the p40 outlier.

For small-molecule strategy the reading is: the clinically winning epitopes are large, receptor-overlapping surfaces centered on the p19 hotspot region — exactly the surface an orthosteric small

Table 2. Verified PDB inventory of IL-23, IL-23R and IL-23-antagonist structures (all verified against the RCSB archive). Codes link to the RCSB entries. Methods: X = X-ray crystallography, EM = cryo-EM. Resolutions in Å.

PDB	Method	Res. (Å)	Content	Source
3DUH	X	2.3	IL-23 (p19+p40), mature p19 numbering	Lupardus & Garcia 2008 [11]
3D87	X	2.9	IL-23	Beyer, JMB 2008
3D85	X	1.9	IL-23 + Fab 7G10 (binds p19 only)	[11]
3QWR	X	3.25	IL-23 + adnectin straddling p19/p40	—
4GRW	X	2.55	IL-23 + nanobodies 37D5 (p19 blocker), 124C4, 22E11 (p40 blocker)	Front Immunol 2017
5MJ3	X	—	IL-23 + alphabody MA12 antagonist (p19)	Nat Commun 2014
5MXA	X	2.5	Apo IL-23 (unbound reference)	2018 complex paper
5MZV	X	2.8	IL-23 : IL-23R (D1–D3) : nanobody 22E11 chaperone — first cognate receptor complex	Immunity 2018
5NJD	X	3.9	IL-23 + briakinumab Fab (p40 D1–D2 junction, tilted toward p19)	Immunity 2018
6WDQ	X	3.4	Full quaternary signaling complex: IL-23 : IL-23R ECD : IL-12Rβ1 (D1–D2)	[7]
6UIB	X	2.74	IL-23 + cyclic peptide 23-652 at distal helix-A pocket	PLoS One 2020
8CR8	X	2.0	Human IL-23 alone	NSMB 2024
8OE4	EM	3.6	Complete ECD signaling complex (cryo-EM)	NSMB 2024
8UUI	X	2.43	IL-23 + cyclic peptide 23-446 in <i>induced orthosteric pocket</i> on p19 — key druggability datum	[10]
9MFG	X	2.85	IL-23R ECD + anti-IL23R VHH (oral single-domain antibody) bound at the p19-binding face of D1	PNAS 2025

molecule must compete at — while the CD loop (S114–Q123) is a conserved, receptor-distal anchor point shared by every successful p19 antibody.

4.5 Induced pockets and druggability

Classically, the p19:IL23R interface would be declared undruggable: it is small ($\sim 900 \text{ Å}^2$), flat, and carries no pre-formed small-molecule pocket in the apo structures ([5MXA](#), [8CR8](#)). Three findings change that assessment.

First, the binding energy is hotspot-concentrated. In the hGH:hGH-receptor paradigm, a small cluster of residues carries most of the interface free energy [4]; the p19 interface has exactly that character — Trp156 dominates, with the ordered AB loop and helix-D N-terminus as secondary contributors — and interfaces where hotspots coincide with a pre-existing or inducible pocket are the tractable cases ([Clackson et al., JMB 1998](#)). The canonical proof that a hotspot-clustered small molecule can displace a cytokine–receptor interaction is the fragment-assembly IL-2 inhibitor [3].

Second, an induced orthosteric pocket exists. Cyclic peptide 23-446 binds p19 in a pocket wedged between the CD loop (109–121) and helix D (159–166) ([Fig. 6](#)), a pocket *cryptic in the apo structures* but opened on peptide binding; the ligand sterically blocks the IL-23R face. Key contacts: the peptide Trp18 forms a π -cation interaction with p19 Arg162, with hydrogen bonds to the L116 backbone, Ser46 and Ser119; SAR optimization reached low-nanomolar affinity [10]. The same study validated a fluorescence-polarization probe and HTS assay and ran a pilot fragment-library screen against the pocket (319 hits, 15 fragments with $\geq 50\%$ inhibition) — the enabling assay and hit set for a true small-molecule campaign. A second, distal pocket at the helix-A N-terminus

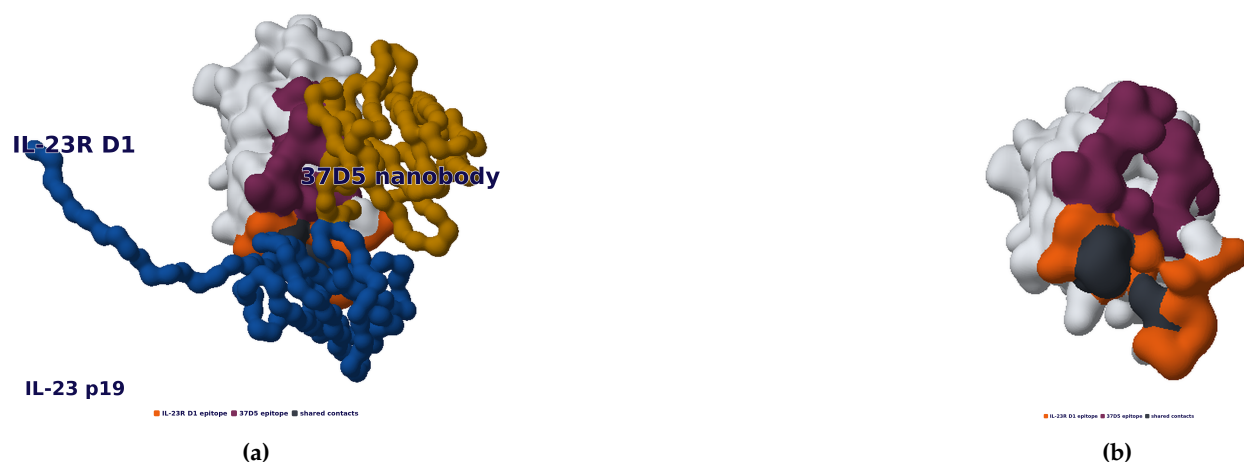


Figure 4. Epitope overlap between the receptor and a p19-blocking binder. (a) p19 surface in white/gray with the IL-23R D1 epitope in orange and the epitope of nanobody 37D5 ([4GRW](#), a p19-blocking nanobody) in plum; shared contacts in dark gray. IL-23R D1 is drawn in blue and the nanobody in gold. (b) p19 alone, same colour mapping, viewed down the receptor-binding face. The 37D5 epitope covers p19 residues 43–49, 112–123 and 156–163 and overlaps the IL-23R footprint including the Trp156 hotspot — a structural explanation of how a protein binder sterically occludes receptor engagement. Rendered from [4GRW](#) and [5MZV](#).

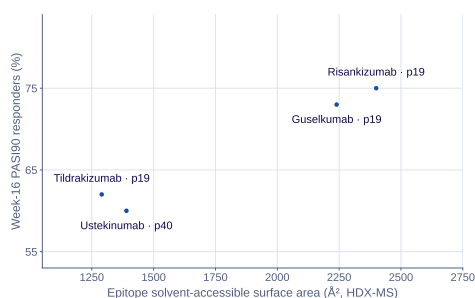


Figure 5. Antibody epitope size versus clinical response. Epitope solvent-accessible surface area (from HDX-MS epitope mapping; approximate) versus week-16 PASI90 response for the four antibodies mapped. This is an association across $n = 4$ antibodies (from the 2024 HDX-MS study; not a dose–response relationship), shown without a fitted line. Ustekinumab binds the shared p40 subunit, not p19, and is retained only as an outlier reference point. No co-structure exists for any of the marketed p19 antibodies — all epitopes are HDX-MS-derived.

(UniProt residues 19–32) is occupied by cyclic peptide 23-652 at low micromolar affinity and is also IL-23R-inhibitory (6UIB).

Third, the receptor side is separately druggable: an engineered, protease-stable oral single-domain antibody binds the p19-binding face of IL-23R D1 (residues 25–31, 56, 98–102, 108–118) in 9MFG, i.e. orthosteric competition on the receptor is achievable with a gut-stable protein scaffold. No experimental small-molecule co-structure exists for either p19 or IL-23R; the few published computational virtual-screening studies against these surfaces are exploratory only, and we treat them as hypothesis generators.



Figure 6. Druggability of the p19 surface. (a) p19 surface in gray with IL-23R-contact residues in orange and residues contacted by cyclic peptide 23-446 in plum (from 8UUI). The peptide-contacted patch sits immediately adjacent to the receptor-contact patch, wedged between the CD loop (109–121) and helix D (159–166). (b) Close-up of the same region with p19 as a blue cartoon, showing the near-complete overlap of the IL-23R contact patch (orange) and the 23-446 contact residues (plum): a ligand in this induced pocket sterically blocks receptor engagement. Rendered from 8UUI [10].

5 Discovery strategies for small molecules

5.1 What the precedents teach

For two decades, class I/II helical cytokine–receptor interfaces (~ 600 – $1,500$ Å², featureless) were considered beyond small molecules. That generalization is disproven several times over, always by the same logic: compete at a *clustered* set of hotspots, or bind a pocket that occludes them. The precedents that matter here: icotrokinra (oral macrocyclic peptide IL-23R antagonist; Section 3); the IL-2 fragment-assembly inhibitor, the classical small-molecule displacement of a cytokine–receptor interaction [3]; DEL-derived small-molecule IL-17A inhibitors optimized to in-vivo-active compounds (Ramos et al., J Med Chem 2024); DEL-derived low-molecular-weight IL-36R antagonists competing at a class-I cytokine receptor interface [13]; structure-guided IL-17A macrocyclic peptides with solved complexes (5HI3–5HI5, Sci Rep 2016); and IL-23 itself, with display-derived peptides 23-652 (6UIB) and 23-446 (8UUI) carrying solved complexes and SAR [10]. The counterpoint: an integrated hit-identification assessment of IL-1 β documents how hard cytokine pockets remain even when they exist (J Med Chem 2024) — designed to fail fast.

5.2 Strategy taxonomy

Table 3 organizes the approaches with their pros and cons. Our portfolio logic: *lead* with a peptide/macrocyclic program against the p19 pockets or the IL23R D1 face (the icotrokinra-validated

Table 3. Strategy taxonomy for disrupting the IL-23 p19:IL23R interaction, with precedents, advantages and liabilities.

Strategy	Precedent	Pros	Cons
Orthosteric small molecule in the p19 induced pocket	8UUI pocket + validated fragment screen [10]; IL-2 inhibitor [3]; DEL-derived IL-17A/IL-36R molecules [13]	Full oral ADMET; titratable PK; hotspot-coincident pocket exists	Flat interface; high potency bar (sub-10-nM vs ~27 pM assembly); pocket is ligand-induced
Allosteric / cryptic pocket on p19 or IL-23R, away from the interface	Mixed-solvent MD (CrypticScout , 2020)	Selectivity; no direct competition with nM binding	Pockets must be induced; occupancy-function link uncertain; needs MD + fragment validation
Peptide / macrocycle mimicking IL-23R D1 or the p19 epitope	Ictrokinra (approved, reporting level); alpha-body 5MJ3 ; display peptides 6UIB ; oral VHH 9MFG	Covers large epitopes with hotspot clustering; nM-pM affinity reachable; clinically validated	Proteolysis/permeability engineering; IP-crowded; peptide COGS
DNA-encoded / display libraries against p19 or IL-23R	DEL-derived IL-17A and IL-36R hits (J Med Chem 2024; [13]); IL-23 display peptides (6UIB)	Proven at cytokine interfaces; huge library sizes	Requires soluble target; affinity-maturation burden
PROTAC / molecular glue vs IL23R or IL23A	No published IL-23 degrader known	Catalytic potency ceiling	Extracellular/secreted targets are poor degrader substrates; LYTAC-class only; exploratory
Indirect: TYK2 (JH2) or STAT3 downstream blockade	Deucravacitinib ; zasocitinib (TAK-279, JAMA Dermatol 2024)	Oral, validated, chemistry exists	No cytokine selectivity (TYK2 also mediates IL-12/IFN); STAT3 oncology-grade; ceiling below IL-23R blockade

modality, noting a crowded patent estate), *run in parallel* a cryptic-pocket small-molecule track on p19/IL23R using mixed-solvent MD and fragment validation, and treat downstream TYK2/STAT3 chemistry as a combination or partnering option rather than a proprietary strategy — it lacks cytokine selectivity and is already crowded.

5.3 Enabling computational and assay platform

Hotspot identification (alanine scanning, fragment-docking hotspot mapping via FTMap, cryptic-site prediction via CryptoSite) feeds into mixed-solvent molecular dynamics for cryptic-pocket detection, available as a validated workflow ([JCIM](#) 2018) and as the PlayMolecule CrypticScout implementation ([JCIM](#) 2020; extension [CrypToth](#), [JCIM](#) 2025), directly runnable on p19 (apo [5MXA](#)/[8SCR8](#), bound [8UUI](#)) and IL-23R D1 ([9MFG](#)). Ensemble docking into hotspot-overlapping pockets followed by FEP is the right engine for the small-molecule track; the published fluorescence-polarization probe and fragment screen [10] provide the starting HTS cascade. Structurally, IL-23 crystals are tractable but the signaling complex needed a nanobody chaperone ([5MZV](#)); plan for chaperones in co-crystallography and compare free versus bound structures to map induced order.

5.4 Preclinical assay cascade

The functional readouts are established: IL-23-dependent STAT3 phosphorylation (pSTAT3-Y705) in human Th17 cells; IL-23-responsive STAT3-luciferase reporter cells reconstituted with IL-23R/IL-

Table 4. Proposed preclinical target product profile for an oral IL-23-axis agent. Values marked (approx.) are derived from cross-trial or industry-estimate sources and should be re-anchored before external use.

Parameter	Proposed value	Rationale
Indication	Moderate-to-severe plaque psoriasis (adults); expand to PsA, IBD	Largest IL-23 franchise indication
Line of therapy	First systemic; biologic-naïve plus biologic-attrited	Orals win share at entry
Primary efficacy (wk 16)	PASI75 $\geq 70\%$; PASI90 $\geq 50\%$ (stretch 55–60%)	Must clear deucravacitinib decisively; 50–55% approaches guselkumab-like positioning from an oral
Secondary efficacy	sPGA 0/1 $\geq 45\text{--}50\%$; PASI100 $\geq 20\%$ (stretch)	sPGA as regulatory co-primary; PASI100 as payer expectation
Onset	PASI50 by wk 4; PASI75 by wk 8–12	Deucravacitinib onset is slow ($\sim$ wk 12 plateau)
Maintenance (wk 52)	PASI90 maintained $\geq 45\%$	Durability differentiates versus apremilast-class decay
Safety	No MACE/VTE imbalance; herpes-zoster monitoring; infections $\leq$ placebo + 5–10 pts; $\geq 10\times$ unbound Cmax tox margin	JAK-class MACE/VTE concerns make IL-23R specificity the safety narrative
PK	$F \geq 30\text{--}50\%$; $t_{1/2}$ 6–12 h for once-daily dosing; 10–100 mg QD; food effect $< 2\times$	QD, no lab monitoring — the convenience case
DDI risk	Low CYP3A4 dependence	Polypharmacy population
Candidate properties	MW < 500 ; logP 2–4; solubility $\geq 100 \mu\text{g mL}^{-1}$; Fsp3 > 0.3 ; Caco-2 $> 10 \times 10^{-6}$ cm/s; no PAINS; $\leq 7\text{--}8$ synthetic steps	PPI-inhibitor constraints; peptide binders (icotrokinra is 240 mg QD) accept higher MW

12R β 1, JAK2 and TYK2; downstream IL-17A/IL-22 output in PBMC–keratinocyte co-cultures; and a *mandatory* IL-12 counter-screen (shared IL-12R β 1/TYK2 use) plus counterscreening against IL-23R-independent STAT3 activators. In vivo, the IL-23 minicircle DNA model (psoriasiform dermatitis from intradermal IL-23) and imiquimod in K14-VEGF transgenic mice (PLoS One 2015) are the workhorses; IL-23-injection models with CCR6-dependence characterization extend the panel.

6 Target product profile

Table 4 proposes a preclinical-stage TPP for an oral small-molecule (or peptide-fallback) IL-23R antagonist. The market context: psoriasis therapeutics is approximately a \$25–30B market (2023–24, approximate and directional) forecast to grow to \$45–60B in the early 2030s; biologics list at \$40–70k per year versus \$20–40k for orals, and payer step-therapy favors orals at entry — the commercial case for a true oral IL-23-axis systemic is share capture at the first-systemic line, with moderate-to-severe plaque psoriasis (approximately 2–3M US adults, approximate) as the anchor indication and PsA/IBD as expansions.

7 Competitive landscape

Table 5 summarizes the competitive field. Two framing notes. First, stage and efficacy figures below are approximate: cross-trial PASI comparisons are informal, and several topline (icotrokinra

Table 5. Competitive landscape for the IL-23 axis and adjacent oral systemics (status early 2025–2026, approximate; press-level topline flagged).

Program	Sponsor	Mechanism	Stage	Notes
Icotrokinra (JNJ-2113, ex-PN-235)	Protagonist-J&J	Oral peptide IL-23R antagonist	Phase 3 positive; approval reported 2026 (reporting level)	Wk-16 PASI90 55–57%, superior to deucravacitinib; Phase 2b topline is press-release level. Benchmark to beat PASI75 ~67–68% (approx., topline)
TAK-279 / zasocitinib	Takeda (Nimbus)	Allosteric TYK2	Phase 2b positive; Phase 3 in PsA	PASI75 ~67–68% (approx., topline)
BMS-986202	BMS	Allosteric TYK2	Phase 2	PASI75 ~85% claimed (uncertain, topline)
Deucravacitinib	BMS	TYK2 JH2	Marketed	Incumbent oral benchmark (PASI90 ~28%, approx.)
Apremilast	Amgen	PDE4	Marketed	Lowest oral tier (PASI90 ~7%)
Upadacitinib / JAK orals	AbbVie et al.	JAK1	Phase 2 psoriasis	MACE/VTE class limits
Other oral IL-23R programs	Morphic/Lilly, Takeda	Various	Discovery/GI-focused	No public oral IL-23R small molecule in clinic (unconfirmed)

Phase 2b, TAK-279, BMS-986202) are press-release-level and must be re-anchored to peer-reviewed sources. Second, the only program that competes at the molecular level targeted here is icotrokinra — an oral *peptide* bound to IL23R; no public oral small molecule acts directly on IL-23R or p19 (unconfirmed for private programs), which is precisely the open position a small-molecule program would occupy.

8 Risks, safety and translational considerations

Mucosal immunity and antifungal defense. IL-23 supports gut homeostasis — antimicrobial peptide production, epithelial repair and commensal containment — through ILC3s, Th17 cells, $\gamma\delta$ T cells and gut epithelial compartments. The human-genetics quantification is direct: homozygosity for the protective IL23R R381Q allele associates with chronic mucocutaneous candidiasis (PMID 40685773) — IL-23 is required for antifungal immunity. Heterozygous protection with preserved immunity appears safe — consistent with hundreds of thousands of patients on anti-p19 antibodies without a class-attributed MACE or malignancy signal — but long-term registry data (>5 years) are still maturing.

IL-12/IL-23 selectivity and IBD. IL-17A blockade worsens or unmasks Crohn’s disease (brodalumab is contraindicated in Crohn’s), whereas IL-23(p19)-selective blockade does not carry that liability: ustekinumab (p40) is approved for Crohn’s and UC, mirikizumab (anti-p19) for UC and Crohn’s, and guselkumab for UC and Phase 3-positive in Crohn’s (BioDrugs 2026, PMID 42012779; GALAXI 5-year Crohn’s data, PMID 42065683). The early termination of the briakinumab (anti-p40) Crohn’s program is often cited as evidence that dual IL-12+IL-23 blockade is riskier in IBD, but the exact termination rationale could not be verified from the sources retrieved here — flagged

accordingly. For a p19-side binder, IL-12 cross-talk is structurally capped (site II absent from p19); an IL-23R-side binder blocks IL-23R signaling entirely, exactly as icotrokinra does — clinically validated behavior.

Potency bar and ligandability. A competitive inhibitor of a ~ 27 pM multivalent assembly typically needs sub-10-nM affinity — an effectively unlimited ceiling for antibodies, not for small molecules. The 8UUI pocket mitigates this by coinciding with hotspot-adjacent structure, but the IL-1 β ligandability study is the standing reminder to kill fast if fragment chemistry does not converge.

Modality-specific risks. For the peptide fallback: oral durability, permeability engineering and cost-of-goods (and, for protein scaffolds, immunogenicity); icotrokinra's 52-week safety record de-risks the class concept but not any specific new molecule. For a small molecule, attribute safety to the pathway, not the modality — the TPP safety bar is set by the JAK class history, and IL-23R-specific blockade is the narrative advantage.

Genetics caveat. The R381Q protective mechanism is cytoplasmic (reduced pSTAT3 and receptor stability) and cannot be reproduced structurally by an extracellular ligand; an extracellular program bets that blocking p19:IL23R engagement reproduces the clinically validated pharmacology.

9 Recommended preclinical plan

A staged plan, with explicit kill criteria, runs the validated icotrokinra-like modality in parallel with the small-molecule frontier (Table 6).

1. **Computational triage (months 0–3).** Run mixed-solvent MD (CrypticScout) on apo p19 (5MXA, 8CR8), peptide-bound p19 (8UUI) and IL-23R D1 (9MFG); map hotspots by computational alanine scanning and FTMap; prioritize pockets overlapping the W156/helix-D hotspot cluster and the CD-loop anchor (S114–Q123). Output: a ranked pocket list with fragmentability scores.
2. **Hit identification (months 2–9), three parallel tracks.** (a) Fragment screen against the 8UUI pocket using the published fluorescence-polarization probe assay (precedent: 319 hits, 15 fragments at $\geq 50\%$ inhibition [10]), with SAR-by-NMR for fragment evolution; (b) DEL selections against p19 with IL-23R co-presentation (following the IL-17A/IL-36R precedents); (c) mRNA/phage/yeast display of macrocyclic peptides against p19 and IL-23R D1 (fallback track).
3. **Structural and affinity maturation (months 6–18).** Co-crystallography of hits with nanobody chaperones; ensemble docking plus FEP on the 8UUI pocket series; iterate with pSTAT3-Y705 assays in Th17 cells and the reporter-cell panel with the mandatory IL-12 counter-screen. Gate: sub-100-nM IL-23 blockade with $>100\times$ IL-12 selectivity before in-vivo commitment.
4. **In vivo and IND-enabling (months 12–24).** IL-23 minicircle and IMQ/K14-VEGF mouse models with skin PK/PD (unbound skin exposure covering receptor engagement); tolerability to $\geq 10\times$ projected Cmax; infection monitoring; then the IND-enabling tox package.
5. **Decision gates.** Kill the small-molecule track fast if fragment-derived chemistry stalls above low-micromolar IL-23 blockade (IL-1 β ligandability precedent); promote the macrocycle track on the same assays, where the clinical benchmark and the safety profile are already established.

Table 6. Condensed preclinical plan. Assays: pSTAT3-Y705 in human Th17 cells; STAT3-luciferase reporter cells reconstituted with IL-23R/IL-12R β 1/JAK2/TYK2; IL-12 counter-screen mandatory. Models: IL-23 minicircle DNA; imiquimod in K14-VEGF mice.

Phase	Key activities	Go / no-go criteria
0. Computational triage (0–3 mo)	CrypticScout mixed-solvent MD on 5MXA , 8CR8 , 8UUI , 9MFG ; alanine scanning + FTMap hotspot maps	At least one druggable pocket overlapping hotspot cluster on either protein face
1. Hit ID (2–9 mo)	Fragment screen (FP probe), SAR-by-NMR; DEL selections; display macrocycles	Confirmed hits with measurable IL-23 blockade; fragments with clear SAR entry points
2. Maturation (6–18 mo)	Co-crystals with chaperone nanobodies; FEP campaign on 8UUI pocket series; reporter cascade	Sub-100-nM IL-23 blockade, >100 $\times$ IL-12 selectivity, cellular pSTAT3 engagement
3. In vivo (12–24 mo)	IL-23 minicircle; IMQ/K14-VEGF; PK/PD and skin exposure; tox margin	Dose-dependent skin clearance consistent with TPP; $\geq 10\times$ safety margin

Bottom line. The biology is validated down to a protective human allele; the interaction is orally druggable (icotrokinra); and a structure-validated orthosteric pocket on p19 with a running fragment screen [10] offers a credible small-molecule entry point. The proportionate strategy is two-track: small-molecule discovery against the induced p19 pocket and cryptic pockets on both faces, with a macrocyclic-peptide track as the clinically validated fallback — differentiated from icotrokinra by modality economics, not biology.

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